

7

dashboard polish

Make it Look the Part

turn a working sheet into a dashboard Priya is happy to be seen with

Yesterday you built the analytical layer — pivots that answer questions, charts that visualise them, slicers that filter the lot with one click. The numbers are right. The layout, frankly, is not. Today is the craft day.

Number formatting (so 0.31196 reads as 31.2%). The KPI tile (one big number, one small label, Merge & Centre). Bucket grouping (so tenure becomes 0-6 months / 6-12 / 12-24 / 24+). Colour discipline (three colours, applied consistently). Chart polish (remove the noise, leave the signal). And the layout that ties it all together — **the Polish Pyramid**.

And one bigger idea worth pausing on first: what you've built across the week isn't just *analysis*. It's *automation*. We'll start there.

Working sheet vs polished dashboard — what's the actual difference?

— you, comparing your sheet to a professional one

Mostly: number formatting, colour discipline, and an eye for layout.

all teachable. all today.



BRAIN POWER

You've seen good dashboards and bad ones. Think of one of each. What's the *single* visual thing that made the bad one look unprofessional? Note it down. Odds are this chapter fixes it.

turn over — and reframe what you've built →

The craft day

Pivots and charts are on the sheet. Numbers are right. Today the work is making it look like the kind of dashboard a senior leader is happy to share in a meeting.

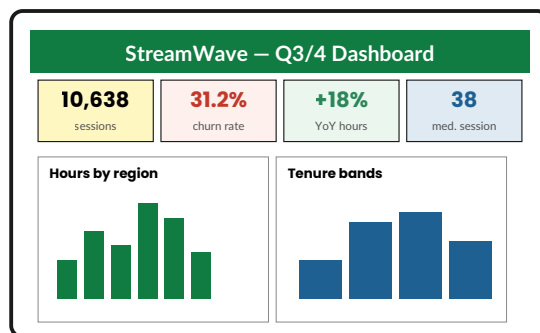
POLISH CHECKLIST

- number formats
- KPI tiles
- bucket grouping
- colour discipline
- chart polish

Polish Pyramid
ties it together

Priya:

"This is going to
land in the exec
meeting. Make it
look professional"



It's not a report — it's automation

Before we dive into formatting, pause on what's now sitting on your screen. You've been calling it "a dashboard" or "the analysis". A senior leader would call it something else: **an automated reporting pipeline**. Two pieces of automation, working together.

Automation #1 — Power Query & Refresh

Every cleaning step, every merge, every key standardisation lives as a *recorded transformation* in Power Query. When next month's data lands, you don't redo the work — you press **Data → Refresh All** and the entire pipeline reruns in under thirty seconds. That's the automation of the *data layer*.

Automation #2 — PivotTables

A PivotTable is not a one-off summary. It's a *configured aggregation* that recalculates whenever its source data changes. You set up "sessions by region by plan type" once; from that point onwards, the PivotTable answers that question forever — including next month, next quarter, next year. That's the automation of the *analytical layer*.

Together they form the toolkit

The two pieces hand off to each other on every Refresh. Source files → Power Query reruns → Tables update → PivotTables rebuild → PivotCharts redraw → KPI cells recalculate. One button press; the whole stack stays current. That's what makes the work you've done this week **recurring value** rather than a one-time deliverable.

ASSESSOR FAVOURITE

Asked "what's the difference between what you've built and a normal Excel report?", the one-sentence answer: "This is an automated pipeline, not a one-off report. Power Query Refresh keeps the data layer current and PivotTables keep the analytical layer current. Both run on a single button press — so next month the same dashboard reflects next month's data with zero rework." That reframe — from "report" to "automated pipeline" — is distinction-band language.

Number formatting

A churn rate that reads `0.31196` isn't a churn rate — it's a maths error waiting to happen. Number formatting is what turns the raw value into something a human can scan in half a second.

The dialog you'll use most

Select the cell(s) → **Ctrl+1** (or right-click → Format Cells) → **Number tab**. Pick the category, set decimal places, tick the thousands separator if useful. Click OK.

The formats you'll need today

Raw	Format	Displays as
<code>0.31196</code>	Percentage, 1 decimal	<code>31.2%</code>
<code>10638</code>	Number, thousands sep	<code>10,638</code>
<code>0.184</code>	% with +/- sign	<code>+18%</code>
<code>15/09/2024</code>	Date <code>dd mmm yyyy</code>	<code>15 Sep 2024</code>

Custom number formats — useful tricks

In Format Cells, pick **Custom**. The format string controls how the number renders. The semicolons separate the formats for positive / negative / zero. Useful patterns:

- `0.0%` — percentage to one decimal.
- `#,##0` — thousands separator, no decimals.
- `[Red]-0%; [Green]+0%` — negatives red, positives green. Great for variance KPIs.
- `#,##0,"k"` — divides by 1,000 and appends "k". `10638` renders as `11k`.

SHOW THIS OFF

When you apply a custom format, narrate it: *"I'm formatting variance as red for negative, green for positive — so Priya knows at a glance which direction it moved without reading the number first."* Format choices are *design* choices; saying that out loud is what marks the difference between a data tech and a spreadsheet user.

The KPI tile

The four boxes across the top of a dashboard — the ones with a big number and a small label underneath — are the first thing the eye lands on. They're called *KPI tiles*, and they're built from one Excel feature: **Merge & Centre**.

Build a single KPI tile — step by step

1. Pick a clean area of your sheet — say cells **B4:D7**. Four columns wide, four rows tall (gives you room for both the number and the label).
2. Select **B4:D5** (the top half, two rows). **Home → Merge & Center**. The three cells become one big cell.
3. In that merged cell, type a formula that returns your KPI value. e.g.
`=COUNTA(tblSubs[subscriber_id])` for total subscribers.
4. Format the cell: **Home → Font Size 28, Bold**, set the number format you want (e.g. thousands separator).
5. Select **B6:D7** (the bottom half). Merge & Center again. Type the label — e.g. *total subscribers* — at font size 10, light grey.
6. Select the whole **B4:D7** block. Apply a fill colour (light yellow for headline KPIs, light red for warnings, light green for positives). Add a thin border (Home → Borders → Outside Borders).

Repeat across the top of the sheet for each KPI. Four tiles is the usual upper limit before they start fighting for attention.



DON'T MERGE & CENTRE INSIDE A TABLE

Merge & Centre breaks sorting and filtering inside an Excel Table — the structure expects one value per cell. Only use Merge & Centre on the *dashboard sheet*, where your KPI tiles live, never on the Tables themselves. Keep the data structures clean and the presentation layer separate.

Bucket grouping

On Day 6 you grouped dates by month and quarter — that's the date version of bucket grouping. Power Query's PivotTable can do the same thing with any *numeric* field: group a continuous value into bands. Hugely useful for segmentation.

A worked example — tenure bands

`tenure_months` is currently a number from 0 to about 70. As a PivotTable row label, that's 70 distinct rows — useless for a chart. Bucket it into bands:

1. Drop `tenure_months` into the **Rows** drop zone.
2. Right-click any value in the row labels → **Group...**
3. In the Grouping dialog set **Starting at:** 0, **Ending at:** 72, **By:** 6. Click OK.
4. The pivot collapses from 70 rows to 12 — each labelled `0-5`, `6-11`, `12-17` etc. Rename the row label to something readable like `tenure_band`.

Custom buckets — when the dialog isn't flexible enough

The Group dialog assumes equal-width buckets. For unequal bands (e.g. 0-6 / 6-12 / 12-24 / 24+), build a helper column on the Subscribers Table first using a Conditional Column back in Power Query, or with `=IFS` in Excel:

```
=IFS([@tenure_months]<6, "0-6 months", [@tenure_months]<12, "6-12 months",  
    [@tenure_months]<24, "12-24 months", TRUE, "24+ months")
```

Then drag that new column into the pivot's Rows instead of the raw `tenure_months`. PivotTables sort alphabetically by default — for the order above, use a "Sort By" column or just live with the alphabetical order (0-6, 12-24, 24+, 6-12) and rename for human eyes.

SHOW THIS OFF

After bucket-grouping `tenure_months`, narrate: *"I bucketed tenure into 6-month bands. The raw column had 70 distinct values — useless for a chart. Twelve bands shows the pattern at a glance and matches how Priya talks about tenure cohorts."* Choosing the bucket size *is* the analysis decision.

Colour discipline

The number-one tell of a homemade dashboard is too many colours. The number-one tell of a professional one is restraint. Three or four colours, used *consistently*, will beat a rainbow every time.

Pick a palette and define what each colour means

Three or four hex codes, used semantically (the same colour always means the same thing). For an analytical dashboard:

Colour	Meaning	Where you'd use it
 Dark blue	Neutral / primary data	Default chart bars, main KPI
 Green	Positive / good direction	YoY growth, retention
 Red	Negative / warning	Churn, errors, things to fix
 Grey	Background / context	Gridlines, labels, supporting text

Two warnings worth knowing

- **Colour-blindness.** About 8% of men can't distinguish red and green reliably. Never use red and green as the *only* signal. Pair colour with a shape, icon, or text label so the dashboard still reads in greyscale.
- **Brand vs analytical.** If StreamWave had brand colours (a specific blue, a specific orange), use them for the dashboard *title and chrome* but stay disciplined with red/green/grey for the data layer. The brand stays consistent; the data stays readable.

ASSESSOR FAVOURITE

Asked "why these colours?", the one-sentence answer: "Three semantic colours — dark blue for neutral data, green for positive, red for negative. Always the same colour for the same meaning across the whole dashboard. Anything else would force Priya to learn a colour code that doesn't exist anywhere else."

Conditional formatting in depth

Conditional formatting was introduced briefly on Day 6. Today is the proper version — when each of the four families is the right tool, with examples for the StreamWave dashboard.

Data Bars — for ranked summary tables

Tiny bars inside cells, length proportional to the value. Best on a single-column summary table where the eye wants to compare lengths. **Home → Conditional Formatting → Data Bars → Gradient Fill**. Pick a colour matching your palette. *Use for:* top-10 most-watched titles (Hours column with data bars makes the ranking obvious without a chart).

Colour Scales — for matrix-style pivots

Heat-map gradient across a 2D range. Low values one colour, high values another. **Conditional Formatting → Color Scales**. *Use for:* a PivotTable showing churn rate by region × plan_type — the cells light up where churn is worst.

Icon Sets — sparingly

Traffic lights, arrows, flags. Use icon sets when the categorical meaning is more important than the precise value. **Conditional Formatting → Icon Sets**. *Use for:* month-over-month trend column (up arrow / flat / down arrow). Don't apply icons to every column — they get noisy fast.

Highlight Cells Rules — for thresholds

Apply a colour when a cell crosses a threshold. **Conditional Formatting → Highlight Cells Rules → Greater Than...** (or any of the other comparisons). *Use for:* the churn rate KPI cell — turns red if it climbs above 30%, automatically, on every Refresh.

SHOW THIS OFF

When demonstrating conditional formatting, point at the *auto-update* behaviour: *"This colour rule lives on the cell, not on the value. So when next month's data comes in and the churn rate changes, the colour follows automatically — same Refresh, same one button press, the dashboard is still self-maintaining."* The link between Refresh and conditional formatting is what makes it automation, not decoration.

Slicer styling

Default slicers are functional but ugly — chunky buttons, mismatched colours, awkward spacing. Three settings and they go from "Excel out of the box" to "part of the dashboard."

Slicer settings — buttons that fit

1. Click the slicer to select it.
2. **Slicer → Buttons → Columns** (on the ribbon). Increase from 1 to 2 or 3 if your slicer has many items — fits more in the same vertical space.
3. **Slicer → Buttons → Height** and **Width**. Set both to round numbers (e.g. 0.3" height, 1.0" width) so the buttons line up cleanly.
4. **Slicer → Size**. Manually set the slicer's overall height and width so it fits the slot in your layout — typically 1.5" wide, however tall it needs to be.

Slicer styles — match your palette

The slicer's colour scheme is on the Slicer ribbon, under **Slicer Styles** (a gallery of small previews). Pick one that uses your dashboard palette — the dark-blue default usually works well, or duplicate a style and recolour it. To set a custom colour: right-click the style → Duplicate → adjust the fill/border colours → save with a new name.

Connect to multiple PivotTables

Already mentioned on Day 6 but worth repeating because it's the slicer's superpower: **right-click slicer → Report Connections...** → tick every PivotTable you want it to filter. One click on the slicer now filters the whole dashboard.

SHOW THIS OFF

When you size the slicer, narrate the choice: *"I set the slicer to two columns and reduced the button height — fits eleven regions in the same space the default takes for four. Layout is part of the analysis."*

Chart polish

A default Excel chart contains four pieces of *chart junk* — visual stuff that adds nothing and distracts. Removing them takes one minute per chart and lifts the dashboard out of the spreadsheet-looking-thing category.

The four bits of junk to remove

1. **Unnecessary gridlines.** Click any horizontal gridline → press Delete. Almost no chart needs horizontal gridlines. Vertical ones are even rarer.
2. **Redundant axes.** If you've added data labels to your bars, the value axis is now redundant — delete it. Same chart, less ink, easier to read.
3. **The default chart title.** Excel adds "Chart Title" — click it, type the actual title (e.g. "*Hours by region — Q3 2024*"). Or delete it entirely if your dashboard has a higher-level title above the chart.
4. **The default colour scheme.** Click any bar → format the data series → fill colour = your palette's blue (or green/red where semantic). Apply consistently across all charts. Greatest single change in dashboard polish.

What to add — sparingly

- **Data labels** on the bars/columns themselves — eliminates the need for the value axis.
- **A short caption** beneath the chart in a 9pt grey font — gives context without crowding the title.
- **Reference lines** (e.g. industry average, prior year) — adds meaning to the comparison.

ASSESSOR FAVOURITE

When demonstrating chart polish, narrate the *removal*: "*I deleted the gridlines and the value axis because the data labels make them redundant. Same data, less ink, the comparison is what stands out.*"
Less is more in chart design — saying that out loud is the distinction-band move.

Halfway House

Four scenarios. Pick the right polish move for each. Answers below the line; wrong answer = the page you should re-read.

1. Your churn rate cell shows 0.31196. You want it to read 31.2%.

- ☐ Merge & Centre ☐ Number Format → Percentage ☐ Conditional Formatting
☐ Bucket Group

2. The tenure_months column has 70 distinct values. You want a chart with about 12 tidy bands.

- ☐ Merge & Centre ☐ Number Format ☐ Conditional Formatting ☐ Bucket Group

3. You want the churn-rate KPI cell to automatically turn red when the value exceeds 30%.

- ☐ Merge & Centre ☐ Number Format ☐ Conditional Formatting → Highlight Cells ☐ Slicer Style

4. The four KPI tiles at the top of the dashboard need to look uniform – one big number, one small label each.

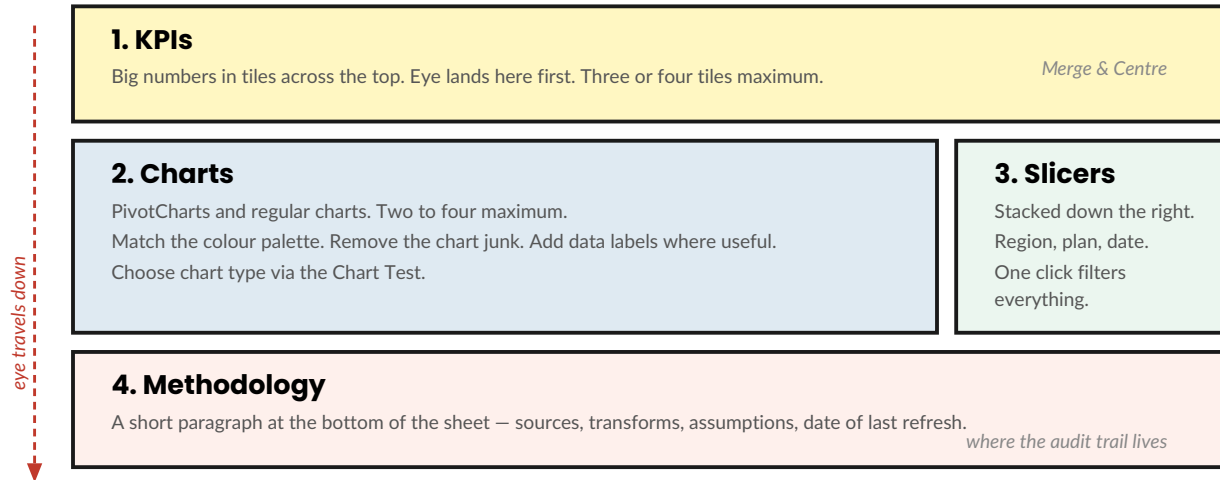
- ☐ Merge & Centre ☐ Bucket Group ☐ Icon Set ☐ Conditional Formatting

Answers – check yourself

1. **Number Format → Percentage.** Ctrl+1, Percentage, 1 decimal place. Done.
2. **Bucket Group.** Right-click any value in the row labels → Group → set Starting/Ending/By.
3. **Conditional Formatting → Highlight Cells.** Greater than 0.30 → Light Red Fill. Updates on every Refresh automatically.
4. **Merge & Centre.** Pair of merged regions per tile – top for the number, bottom for the label – same shape across all four.

The Polish Pyramid

Four layers, top to bottom. Same shape on every dashboard you'll build for the rest of your career. The reader's eye lands on the top-left and works outward; the layout has to match that path.



One sheet. No scrolling. The reader takes about ten seconds to absorb the top KPI tiles, twenty seconds on the charts, and a glance at the methodology if they care how it was built. That's it — three quick passes and the story has landed.

Documentation & GDPR

A polished dashboard you can't explain — or that exposes data it shouldn't — fails on a different axis. Two short topics: where your audit trail lives, and the three GDPR rules you need by heart.

Where the audit trail lives

- **Applied Steps** in each Power Query — the full transformation history. Rename steps in plain English before handing over.
- **Source files listed** in the methodology cell at the bottom of the dashboard.
- **Table-column references** (`tblActivity[duration]`) — formulas that read like sentences.
- **A methodology note** — three or four sentences covering sources, transforms, assumptions, and "last refreshed" date. Enough for a colleague to pick up where you left off.

GDPR — three rules for any dashboard with personal data

Your StreamWave data contains personal data: names, regions, plan types, viewing history. UK GDPR applies. Three practical rules:

- **Report on aggregates, not individuals.** *"31.2% churn rate"* is fine. *"Isla Hussain cancelled on the 19th"* is not — unless there's a specific business reason and the dashboard is locked to the right audience.
- **Minimise what you include.** The Subscribers Table can live as Connection Only; only aggregated summaries need to be on a visible sheet. Don't load personal details to a sheet you don't need to.
- **Be careful where the file goes.** Personal data shouldn't be emailed to *"all employees"* or posted to a general channel. Restrict access to people with a business reason.

ASSESSOR FAVOURITE

The data ethics question is asked almost every assessment. The one-sentence answer: *"This data is personal — names, regions, viewing history — so I've reported on aggregates rather than individuals, kept the raw Subscribers Table as Connection Only, and restricted file access to people with a business reason to see it."* Three practical actions, not a lecture on GDPR.

The pre-share QA pass

Before the dashboard leaves your screen — before you email it to Priya, before the exec meeting — run a ten-point QA pass. Two minutes. Catches almost every embarrassing miss.

- ☐ Every headline number has been verified with **The Two-Route Check** at least once — note in methodology cell.
-
- ☐ Every chart has a title that explains the finding, not just the data field.
-
- ☐ Every percentage has a % sign; every currency has a £; every count has a thousand separator.
-
- ☐ No `#DIV/0!`, `#REF!`, or `#VALUE!` anywhere — even hidden behind a filter selection.
-
- ☐ Press **Data → Refresh All**. Pivots and KPIs all update without errors.
-
- ☐ Click each slicer to its extremes (all selected, one selected, none selected). Nothing breaks; nothing returns blank where it shouldn't.
-
- ☐ The methodology cell names the source files, the date of last refresh, and any assumptions.
-
- ☐ No personal data shown at individual level on any visible sheet.
-
- ☐ Open in another zoom level (75%, 150%) — layout still holds.
-
- ☐ Save. Close. Reopen. Everything still works — proves it'll work for Priya tomorrow.
-

The first time you'll forget at least three of these. Tape the list to the side of your monitor for the first week of using a new dashboard. After that it's muscle memory.

Bullet Points — the chapter in 30 seconds

- **It's automation, not a report.** Refresh + PivotTables = a recurring system updated on one button press.
- **Number formats first.** A KPI without a format is a maths error waiting to happen.
- **The KPI tile.** Merge & Centre, big number above, small label below. Max four tiles.
- **Bucket grouping** turns 70 distinct values into tidy bands. Use `=IFS` for unequal buckets.
- **Three colours, used semantically** — same colour, same meaning, everywhere.
- **Chart polish = removal.** Gridlines, redundant axes, default titles, default colours — gone.
- **The Polish Pyramid.** KPIs · Charts · Slicers · Methodology, top to bottom.
- **The pre-share QA pass.** Ten-point check before the dashboard leaves your screen — Two-Route on every headline, refresh, slicer extremes, methodology cell complete.

TEST DRIVE

Before closing the laptop, run this self-check:

- ☐ Every number on the dashboard has the right format (% / thousands / decimals / dates).
- ☐ KPI tiles built with Merge & Centre at the top of the sheet.
- ☐ At least one PivotTable uses bucket grouping for a numeric field.
- ☐ Three colours, used consistently. Every chart has had its junk removed.
- ☐ Slicers styled and connected to multiple pivots; methodology paragraph at the bottom.
- ☐ The ten-point QA pass run and every item ticked.
- ☐ You can name the four layers of The Polish Pyramid from memory.

COMING UP — DAY 8

Effective Working Practices. Tomorrow we move off the keyboard and onto the soft skills the EPA grades just as hard. Audience-appropriate communication, stakeholder management, the three voices (exec / peer / new starter), CPD, multi-functional teamwork, inclusive working, and how to write a methodology note that anyone could pick up tomorrow morning. Plus the **Headline → Evidence → Caveat** framework for presenting findings.

that's Day 7 done — the dashboard now looks the part.